Experiment 01: Design of DFA to Accept Even Number of a's

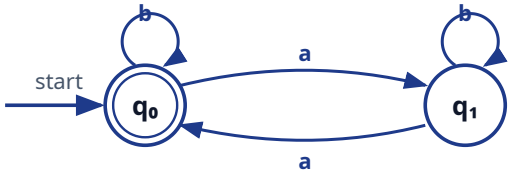
Course: CSA13 – Theory of Computation | Syllabus Ref: Exp 05 | Faculty Evaluation: [/ 10]

1. Aim

To design a Deterministic Finite Automaton (DFA) using an automata simulator to accept all strings containing an even number of 'a's over the alphabet Σ = {a, b}.

2. Theoretical Formulation & Diagram

- **Q:** $\{q_0, q_1\}$; **Start State:** q_0; **Final State Set (F):** $\{q_0\}$ (0 is even).
- $\delta(q_0, a) = q_1$, $\delta(q_0, b) = q_0$; $\delta(q_1, a) = q_0$, $\delta(q_1, b) = q_1$.



3. Simulator Verification

String	Count of 'a's	Simulator Path	Verdict
b	0 (Even)	$q_0 \rightarrow q_0$	Accepted
aa	2 (Even)	$q_0 \rightarrow q_1 \rightarrow q_0$	Accepted
baab	2 (Even)	$q_0 \rightarrow q_0 \rightarrow q_1 \rightarrow q_0 \rightarrow q_0$	Accepted
a	1 (Odd)	$q_0 \rightarrow q_1$	Rejected

4. Result

The DFA accepting even number of a's was designed in the simulator and verified.

Experiment 02: Design of DFA to Accept Odd Number of a's

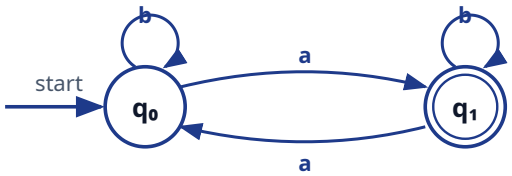
Course: CSA13 – Theory of Computation | Syllabus Ref: Exp 06 | Faculty Evaluation: [/ 10]

1. Aim

To design a Deterministic Finite Automaton (DFA) using an automata simulator to accept all strings containing an odd number of 'a's over the alphabet Σ = {a, b}.

2. Theoretical Formulation & Diagram

- **Q:** $\{q_0, q_1\}$; **Start State:** q_0; **Final State Set (F):** $\{q_1\}$ (odd parity).
- $\delta(q_0, a) = q_1$, $\delta(q_0, b) = q_0$; $\delta(q_1, a) = q_0$, $\delta(q_1, b) = q_1$.



3. Simulator Verification

String	Count of 'a's	Simulator Path	Verdict
a	1 (Odd)	$q_0 \rightarrow q_1$	Accepted ($\in F$)
aaa	3 (Odd)	$q_0 \rightarrow q_1 \rightarrow q_0 \rightarrow q_1$	Accepted ($\in F$)
aa	2 (Even)	$q_0 \rightarrow q_1 \rightarrow q_0$	Rejected ($\notin F$)

4. Result

The DFA accepting odd number of a's was designed in simulator and verified.

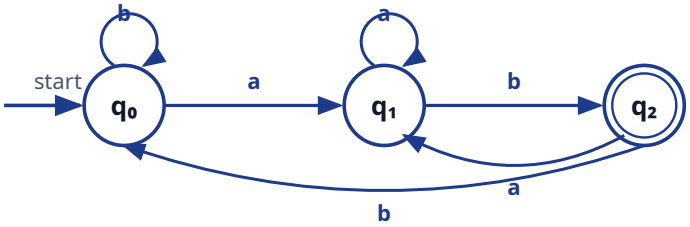
Experiment 03: Design of DFA for Strings Ending with 'ab' over Σ = {a, b} (Test W = aaabab)

Course: CSA13 – Theory of Computation | Syllabus Ref: Exp 07 | Faculty Evaluation: [/ 10]

1. Aim

To design a Deterministic Finite Automaton (DFA) using a simulator to accept all strings ending with 'ab' over Σ = {a, b}, and verify the step trace for **W = aaabab**.

2. Diagram & Transition Function



3. Step Trace for W = aaabab

Step	State	Input	Rule	Next State	Remaining
1	q_0	'a'	$\delta(q_0, a) = q_1$	q_1	aabab
2	q_1	'a'	$\delta(q_1, a) = q_1$	q_1	abab
3	q_1	'a'	$\delta(q_1, a) = q_1$	q_1	bab
4	q_1	'b'	$\delta(q_1, b) = q_2$	q_2	ab
5	q_2	'a'	$\delta(q_2, a) = q_1$	q_1	b
6	q_1	'b'	$\delta(q_1, b) = q_2$	q_2	ε

Decision: Final state is $\mathbf{q_2 \in F}$ ⇒ **W = aaabab is ACCEPTED**.

4. Result

The DFA was designed and the string W = aaabab was verified as accepted.

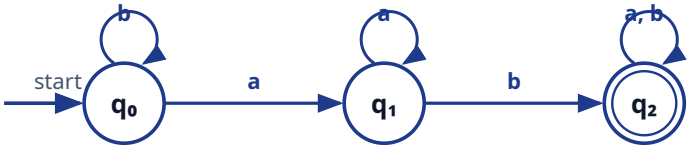
Experiment 04: Design of DFA for Strings Having 'ab' as Substring over Σ = {a, b}

Course: CSA13 – Theory of Computation | Syllabus Ref: Exp 08 | Faculty Evaluation: [/ 10]

1. Aim

To design a Deterministic Finite Automaton (DFA) using an automata simulator to accept all strings having 'ab' as a substring over the alphabet Σ = {a, b}.

2. Diagram & Specifications



3. Test Cases

String	Simulator Path	Verdict
ab	$q_0 \rightarrow q_1 \rightarrow q_2$	Accepted
bab	$q_0 \rightarrow q_0 \rightarrow q_1 \rightarrow q_2$	Accepted
bba	$q_0 \rightarrow q_0 \rightarrow q_0 \rightarrow q_1$	Rejected

4. Result

The DFA recognizing strings containing substring 'ab' was modeled and verified.

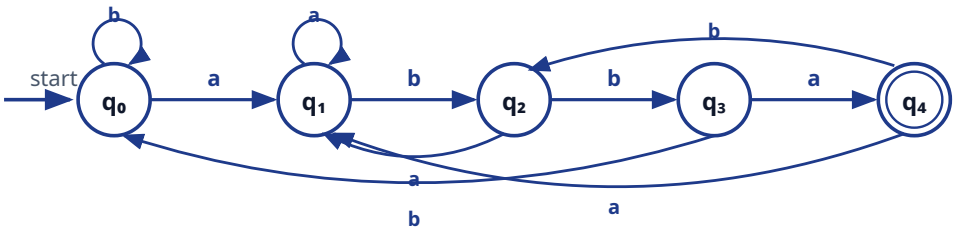
Experiment 05: Draw a DFA for Strings Ending with 'abba' over Alphabet Σ = {a, b}

Course: CSA13 – Theory of Computation | Syllabus Ref: Exp 09 | Faculty Evaluation: [/ 10]

1. Aim

To construct and draw a Deterministic Finite Automaton (DFA) that accepts all strings ending with the pattern 'abba' over the input alphabet Σ = {a, b}.

2. Diagram & Transition Function



3. Verification Traces

String	Transition Sequence	Status
abba	$q_0 \rightarrow q_1 \rightarrow q_2 \rightarrow q_3 \rightarrow q_4$	Accepted

String	Transition Sequence	Status
aabba	$q_0 \to q_1 \to q_1 \to q_2 \to q_3 \to q_4$	Accepted
abbab	$q_0 \to q_1 \to q_2 \to q_3 \to q_4 \to q_2$	Rejected

4. Result

The DFA for strings ending with 'abba' over Σ = {a, b} was constructed and verified.

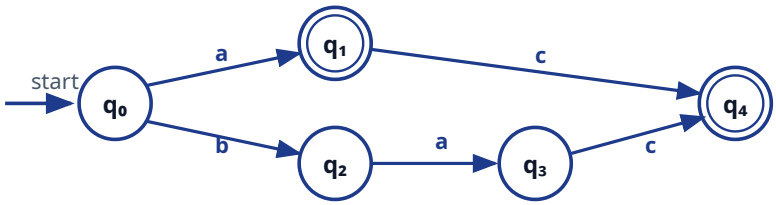
Experiment 06: Design of DFA to Accept Finite Language L = {"a", "ac", "bac"} over Σ = {a, b, c}

Course: CSA13 – Theory of Computation | Syllabus Ref: Exp 12 | Faculty Evaluation: [/ 10]

1. Aim

To design a Deterministic Finite Automaton (DFA) using an automata simulator to accept exactly the finite set of strings {"a", "ac", "bac"} over Σ = {a, b, c}.

2. Diagram & Transition Table



3. Simulator Test Cases

String	Execution Path	Verdict
a	$q_0 \to q_1$	Accepted ($\in F$)
ac	$q_0 \to q_1 \to q_4$	Accepted ($\in F$)
bac	$q_0 \to q_2 \to q_3 \to q_4$	Accepted ($\in F$)
ba	$q_0 \to q_2 \to q_3$	Rejected ($\notin F$)
c	$q_0 \to q_{dead}$	Rejected (Trap)

4. Result

The DFA recognizing finite language {"a", "ac", "bac"} was constructed and tested.

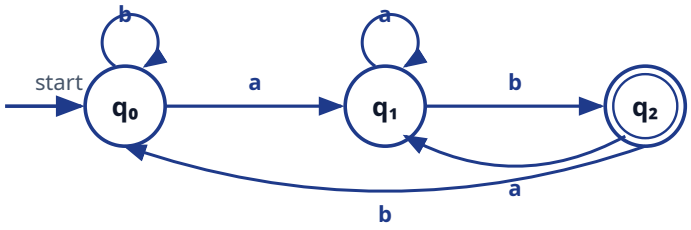
Experiment 07: Design of DFA for Strings Ending with 'ab' over Σ = {a, b} (Trace W = abbaabab)

Course: CSA13 – Theory of Computation | Syllabus Ref: Exp 13 | Faculty Evaluation: [/ 10]

1. Aim

To design a Deterministic Finite Automaton (DFA) using an automata simulator to accept all strings ending with 'ab' over Σ = {a, b}, and verify the execution trace for **W = abbaabab**.

2. Diagram



3. Step-by-Step Simulation Trace for W = abbaabab

Step	State	Input	Rule	Next State	Remaining
1	q_0	'a'	$\delta(q_0, a) = q_1$	q_1	bbaabab
2	q_1	'b'	$\delta(q_1, b) = q_2$	q_2	baabab
3	q_2	'b'	$\delta(q_2, b) = q_0$	q_0	aabab
4	q_0	'a'	$\delta(q_0, a) = q_1$	q_1	abab
5	q_1	'a'	$\delta(q_1, a) = q_1$	q_1	bab
6	q_1	'b'	$\delta(q_1, b) = q_2$	q_2	ab
7	q_2	'a'	$\delta(q_2, a) = q_1$	q_1	b
8	q_1	'b'	$\delta(q_1, b) = q_2$	q_2	ε

Conclusion: DFA halts in accepting state $q_2 \in F \Rightarrow$ **W = abbaabab is ACCEPTED**.

4. Result

The DFA simulation verified that W = abbaabab is accepted.

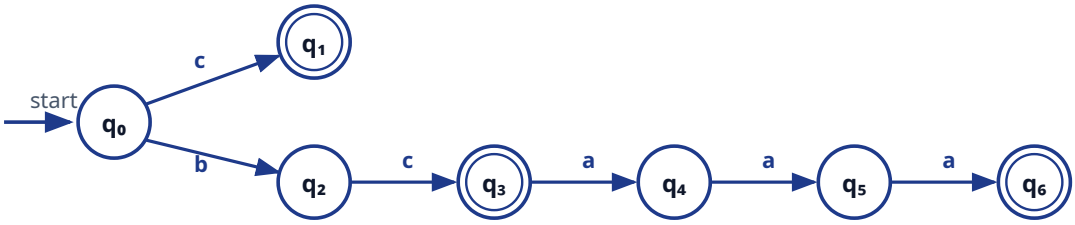
Experiment 08: Design of DFA to Accept Finite Language L = {"bc", "c", "bcaaa"} over Σ = {a, b, c}

Course: CSA13 – Theory of Computation | Syllabus Ref: Exp 14 | Faculty Evaluation: [/ 10]

1. Aim

To design a Deterministic Finite Automaton (DFA) using a simulator to accept the finite language containing input strings "bc", "c", and "bcaaa" over Σ = {a, b, c}.

2. Diagram & Specifications



3. Test Cases

String	State Sequence	Verdict
c	$q_0 \to q_1$	Accepted ($\in F$)
bc	$q_0 \to q_2 \to q_3$	Accepted ($\in F$)
bcaaa	$q_0 \to q_2 \to q_3 \to q_4 \to q_5 \to q_6$	Accepted ($\in F$)
bca	$q_0 \to q_2 \to q_3 \to q_4$	Rejected ($\notin F$)

4. Result

The DFA accepting {"bc", "c", "bcaaa"} was designed and tested.

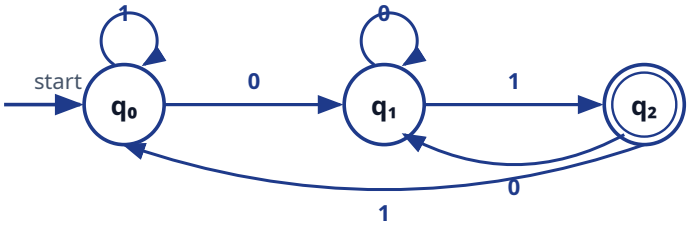
Experiment 09: Draw a DFA for Language of Strings Ending with '01' over Σ = {0, 1}

Course: CSA13 – Theory of Computation | Syllabus Ref: Exp 15 | Faculty Evaluation: [/ 10]

1. Aim

To construct and draw a Deterministic Finite Automaton (DFA) that accepts the language of all binary strings ending with '01' over the alphabet Σ = {0, 1}.

2. Diagram & Transition Function



3. Verification Traces

String	Transition Sequence	Outcome
01	$q_0 \to q_1 \to q_2$	Accepted
1001	$q_0 \to q_0 \to q_1 \to q_1 \to q_2$	Accepted
1010	$q_0 \to q_0 \to q_1 \to q_2 \to q_1$	Rejected

4. Result

The DFA for binary strings ending with '01' was constructed and verified.

Experiment 10: Design of DFA for Strings where a's Always Appear Tripled over Σ = {a, b}

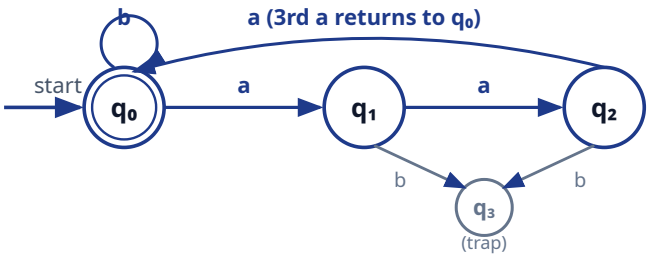
Course: CSA13 – Theory of Computation | Syllabus Ref: Exp 16 | Faculty Evaluation: [/ 10]

1. Aim

To design a Deterministic Finite Automaton (DFA) using a simulator to accept strings in which every occurrence of symbol 'a' appears in consecutive multiples of three ($L = (b + aaa)^*$) over $\Sigma = \{a, b\}$.

2. Diagram & Transition Table

Current State	Input 'a'	Input 'b'	Semantics
$^*\rightarrow q_0$	q_1	q_0	Triplet complete / Safe
q_1	q_2	q_{dead}	1 'a' active (incomplete)
q_2	q_0	q_{dead}	2 'a's active (incomplete)
q_{dead}	q_{dead}	q_{dead}	Violation State



3. Test Cases

String	Transition Sequence	Verdict
b	$q_0 \rightarrow q_0$	Accepted ($\in F$)
aaa	$q_0 \rightarrow q_1 \rightarrow q_2 \rightarrow q_0$	Accepted ($\in F$)
baaab	$q_0 \rightarrow q_0 \rightarrow q_1 \rightarrow q_2 \rightarrow q_0 \rightarrow q_0$	Accepted ($\in F$)
aab	$q_0 \rightarrow q_1 \rightarrow q_2 \rightarrow q_{dead}$	Rejected (Trap)

4. Result

The DFA for language $(b + aaa)^*$ where a's always appear tripled was verified.

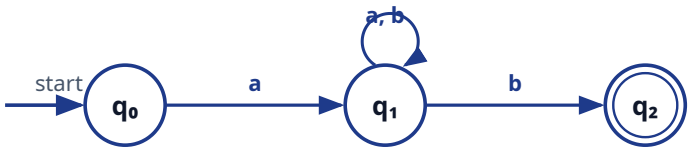
Experiment 11: Design of NFA for Strings Starting with 'a' and Ending with 'b' (Check W = abaab)

Course: CSA13 – Theory of Computation | Syllabus Ref: Exp 17 | Faculty Evaluation: [/ 10]

1. Aim

To design a Non-Deterministic Finite Automaton (NFA) using a simulator to accept strings starting with 'a' and ending with 'b' over Σ = {a, b}, and evaluate whether string **W = abaab** is accepted.

2. Transition Model



3. Evaluation Trace for W = abaab

Step	Active State Set	Input	Next Active State Set
Start	$\{q_0\}$	—	$\{q_0\}$
1	$\{q_0\}$	'a'	$\{q_1\}$
2	$\{q_1\}$	'b'	$\{q_1, q_2\}$
3	$\{q_1, q_2\}$	'a'	$\{q_1\}$
4	$\{q_1\}$	'a'	$\{q_1\}$
5	$\{q_1\}$	'b'	$\{q_1, q_2\}$

Decision: Final active set contains $q_2 \in F \Rightarrow$ **W = abaab is ACCEPTED**.

4. Result

The NFA was simulated and string W = abaab was confirmed to be accepted.

Experiment 12: Design of NFA for Strings Starting and Ending with Different Symbols

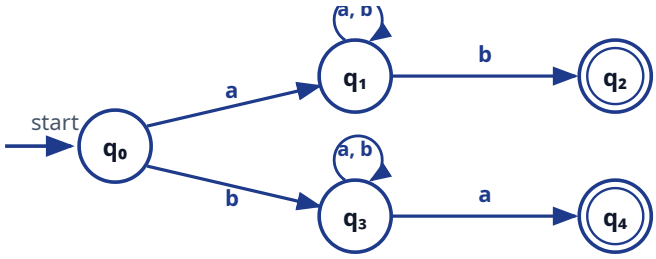
Course: CSA13 – Theory of Computation | Syllabus Ref: Exp 18 | Faculty Evaluation: [/ 10]

1. Aim

To design a Non-Deterministic Finite Automaton (NFA) using a simulator to accept all strings that start and end with different symbols over Σ = {a, b} (Language L = $a(a+b)^*b \cup b(a+b)^*a$).

2. Diagram & Transition Table

State	Input 'a'	Input 'b'
→ q_0	$\{q_1\}$	$\{q_3\}$
q_1	$\{q_1\}$	$\{q_1, q_2\}$
* q_2	∅	∅
q_3	$\{q_3, q_4\}$	$\{q_3\}$
* q_4	∅	∅



3. Test Cases

String	Active State Path	Verdict
ab	$q_0 \to q_1 \to \{q_1, q_2\}$	Accepted ($q_2 \in F$)
ba	$q_0 \to q_3 \to \{q_3, q_4\}$	Accepted ($q_4 \in F$)
abba	Starts 'a', ends 'a'	Rejected

4. Result

The NFA for strings starting and ending with different symbols was verified.

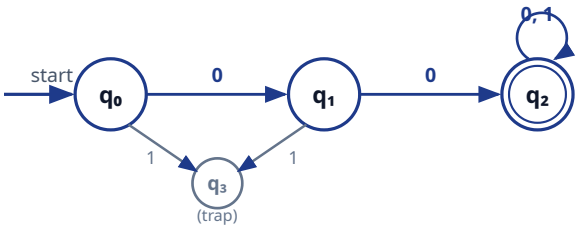
Experiment 13: Draw a DFA for Language of Strings Starting with '00' over Σ = {0, 1}

Course: CSA13 – Theory of Computation | Syllabus Ref: Exp 19 | Faculty Evaluation: [/ 10]

1. Aim

To construct and draw a Deterministic Finite Automaton (DFA) that accepts the language L of all binary strings starting with prefix '00' over Σ = {0, 1}.

2. Model & Diagram



3. Test Cases

String	Execution Path	Result
00	$q_0 \to q_1 \to q_2$	Accepted ($\in F$)
00101	$q_0 \to q_1 \to q_2 \to q_2 \to q_2 \to q_2$	Accepted ($\in F$)
0100	$q_0 \to q_1 \to q_{dead}$	Rejected (Trap)
100	$q_0 \to q_{dead}$	Rejected (Trap)

4. Result

The DFA for strings starting with '00' was constructed and verified.

Experiment 14: Design of DFA for Strings Ending with 'abc' over Σ = {a, b, c} (Trace W = abbaababc)

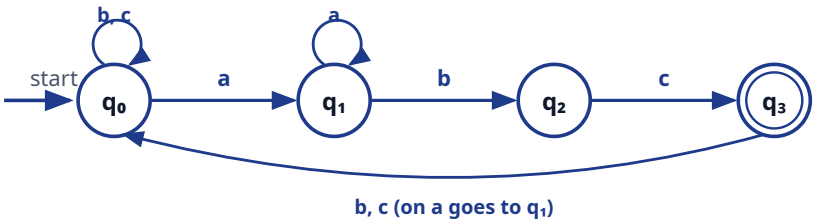
Course: CSA13 – Theory of Computation | Syllabus Ref: Exp 20 | Faculty Evaluation: [/ 10]

1. Aim

To design a Deterministic Finite Automaton (DFA) using an automata simulator to accept strings ending with 'abc' over Σ = {a, b, c}, and verify acceptance for **W = abbaababc**.

2. Diagram & Transition Model

State	Input 'a'	Input 'b'	Input 'c'	Category
→ q_0	q_1	q_0	q_0	Initial
q_1	q_1	q_2	q_0	Matched 'a'
q_2	q_1	q_0	q_3	Matched 'ab'
* q_3	q_1	q_0	q_0	Matched 'abc' (Final)



3. Detailed Simulator Trace for W = abbaababc

Step	State	Input	Applied Transition	Next State	Remaining
1	q_0	'a'	$\delta(q_0, a) = q_1$	q_1	bbaababc
2	q_1	'b'	$\delta(q_1, b) = q_2$	q_2	baababc
3	q_2	'b'	$\delta(q_2, b) = q_0$	q_0	aababc
4	q_0	'a'	$\delta(q_0, a) = q_1$	q_1	ababc
5	q_1	'a'	$\delta(q_1, a) = q_1$	q_1	babc
6	q_1	'b'	$\delta(q_1, b) = q_2$	q_2	abc
7	q_2	'a'	$\delta(q_2, a) = q_1$	q_1	bc
8	q_1	'b'	$\delta(q_1, b) = q_2$	q_2	c
9	q_2	'c'	$\delta(q_2, c) = q_3$	q_3	ε

Decision: Final state is $\mathbf{q_3 \in F}$ ⇒ **W = abbaababc is ACCEPTED**.

4. Result

The DFA for language ending with 'abc' over Σ = {a, b, c} was verified for W = abbaababc.